FYUGP B.Sc. MATHEMATICS HONOURS

ABSTRACT ALGEBRA I (MAT5CJ302)

Semester V · Core in Major

MODULE 1: BINARY OPERATIONS, GROUPS AND ISOMORPHISMS

Model questions on topics the past papers miss

Section A: 1 mark each, answer all

Closure

1 Is the set of odd integers closed under addition? Answer yes or no. **(1 mark)**

Well-defined operations

2 Does $a * b = a/b$ on $\mathbb{Z}$ fail Condition 1 or Condition 2 of a binary operation? Answer 1 or 2. **(1 mark)**

Identity elements

3 Can a set with a binary operation have two different identity elements? Answer yes or no. **(1 mark)**

Group tables

4 Write the group table of a group of two elements $\{e, a\}$ with identity e. **(1 mark)**

Identity elements

5 Is $\langle \mathbb{Z}^+, \max \rangle$ a group? Answer yes or no. **(1 mark)**

Group tables

6 In the group table of a group, how many times does each element appear in each row? **(1 mark)**

One-sided axioms

7 State whether true or false: in a group, a left identity is also a right identity. **(1 mark)**

8 State whether true or false: in a group, a left inverse is also a right inverse. **(1 mark)**

Section B: 2 marks each, answer all

Uniqueness of inverses

9 Show that every element of a group has exactly one inverse. **(2 marks)**

Identity elements

10 Explain why $\langle \mathbb{Z}^+, \max \rangle$ is not a group, although it has an identity element. **(2 marks)**

Group tables

11 Explain why a table in which some element appears twice in one row cannot be the table of a group. **(2 marks)**

Group isomorphisms

12 Write the two conditions a function $f : G_1 \to G_2$ must satisfy to be a group isomorphism. **(2 marks)**

13 Show that $f(x) = \ln x$ is one-to-one and onto from $\mathbb{R}^+$ to $\mathbb{R}$, and that $f(xy) = f(x) + f(y)$. **(2 marks)**

Section C: 4 marks each, answer any two

Groups of functions

14 Let $f_{a,b}(x) = ax + b$ on $\mathbb{R}$ with $a \neq 0$. Show that the composition of two such functions is again of this form, and find the inverse of $f_{a,b}$. **(4 marks)**

Group tables

15 Let $*$ be given on $\{e, a, b\}$ with identity e, and suppose each element appears exactly once in each row and column. Show that $a * a = b$, $a * b = e$, $b * a = e$ and $b * b = a$ are forced. **(4 marks)**

16 Let a be an element of a group G. Show that $x \mapsto a * x$ is one-to-one and onto G, and use this to explain why each element appears once in each row of the table. **(4 marks)**

Section D: 10 marks each, answer any one

Groups of order 4

17 Give two group tables on a four-element set $\{e, a, b, c\}$ that do not define isomorphic groups. Show that they are not isomorphic. **(10 marks)**

One-sided axioms

18 Let G be a nonempty set with an associative operation $*$, a left identity e and, for each a, a left inverse a' with $a' * a = e$. Prove that G is a group. **(10 marks)**

MODULE 1: MODEL ANSWERS

Key steps for checking your work

Section A. 1. No. The sum of two odd integers is even, so H is not closed under addition. 2. Condition 1: a/b is undefined when $b = 0$, so $*$ is not everywhere defined. 3. No. If e and e' are identities then $e = e * e' = e'$ (Theorem 1.15). 4.

$*$	e	a
e	e	a
a	a	e

5. No. $\mathbf{1}$ is an identity ($\max\{a, 1\} = a$), but $\mathbf{2}$ has no inverse, since $\max\{2, x\} = 1$ is impossible. 6. Exactly once: for fixed a, the equation $a * x = b$ has exactly one solution for each b (Theorem 2.17). 7. True. This is the one-sided form of the identity axiom; it follows from the other one-sided axioms. 8. True. The left inverse is also a right inverse, by the one-sided argument in Section 2.

Section B. 9. Let a' and a'' be inverses of a. Then $a' = a' * e = a' * (a * a'') = (a' * a) * a'' = e * a'' = a''$. 10. $\langle \mathbb{Z}^+, \max \rangle$ has identity $\mathbf{1}$, but $\mathbf{2}$ has no inverse. Each element needs an inverse, so it is not a group. 11. If $a * x = a * y$ with $x \neq y$, then left cancellation (Theorem 2.16) gives $x = y$. So no element can repeat in a row. 12. (1) f is one-to-one and onto G_2. (2) $f(a *_1 b) = f(a) *_2 f(b)$ for all $a, b \in G_1$. 13. Bijective: its inverse is $\exp$. Homomorphism: $\ln(xy) = \ln x + \ln y$.

Section C. 14. Composition: $f_{a,b}(f_{c,d}(x)) = a(cx + d) + b = (ac)x + (ad + b)$, with $ac \neq 0$. Inverse: $f_{a,b}^{-1} = f_{1/a,\, -b/a}$. Identity: $f_{1,0}$. Associativity: Theorem 1.13. 15. Row e and column e are forced by the identity. Then $a * a \in \{b, e\}$; $a * a = e$ leads to a repeated b in row b, so $a * a = b$, and the rest follows. The completed table is the group of order 3. 16. Injective: $a * x = a * y$ implies $x = y$ (cancellation). Onto: $x = a' * b$ solves $a * x = b$ (Theorem 2.17). So b occurs once in the row of a.

Section D. 17. Klein group: $a * a = b * b = c * c = e$, with $a * b = c$. Cyclic group $\mathbb{Z}_4$ (with $e = 0$, $a = 1$, $b = 2$, $c = 3$): $a * a = b$. An isomorphism preserves $x * x = e$, so the two cannot be isomorphic. 18. Let $u = a * a'$. Then $u * u = a * (a' * a) * a' = a * a' = u$. If $v * u = e$, then $u = e * u = (v * u) * u = v * (u * u) = v * u = e$. So $a * a' = e$, and $a * e = a * (a' * a) = (a * a') * a = a$. Hence e is an identity and a' is an inverse.

Mapping to the syllabus and textbook

Syllabus item	Topic	Fraleigh 8th ed. section	Model questions
1	Binary operations: closure, well-definedness	1.4, 1.6, 1.26	1, 2
1	Identity and inverses in a binary structure	1.14, 1.15, 1.17	3, 5, 10
3–4	Group isomorphisms	2.22, 2.23	12, 13
5	Groups, one-sided axioms, groups of functions	2.1, 2.14, Exercises 2	7, 8, 14, 18
6	Uniqueness of inverses; cancellation; unique solutions	2.16, 2.17, 2.18	9, 11, 16
6	Finite groups and group tables	2.19, 2.20, 2.21, p. 27	4, 6, 15, 17